python indexing & slicing

In [1]:
```
s1 = 'welcometeam'
s1
```
Out[1]: 'welcometeam'

In [2]:
```
type(s1)
```
Out[2]: str

In [3]:
```
len(s1)
```
Out[3]: 11

In [4]:
```
s1
```
Out[4]: 'welcometeam'

In [5]:
```
s1[1] # forward index
```
Out[5]: 'e'

In [6]:
```
s1
```
Out[6]: 'welcometeam'

In [7]:
```
s1[-1]
```
Out[7]: 'm'

In [8]:
```
s1
```
Out[8]: 'welcometeam'

In [9]:
```
s1[:]
```
Out[9]: 'welcometeam'

In [10]:
```
s1[7:11]
```
Out[10]: 'team'

In [11]:
```
s1
```
Out[11]: 'welcometeam'

In [12]:
```
s1[7:]
```
Out[12]: 'team'

In [14]:
```
s1[:]
```

Out[14]: 'welcometeam'

In [15]:
```
s1[2:10]
```

Out[15]: 'lcometea'

In [16]:
```
s1
```

Out[16]: 'welcometeam'

In [17]:
```
s1[:10]
```

Out[17]: 'welcometea'

In [18]:
```
s1
```

Out[18]: 'welcometeam'

In [19]:
```
s1[4:]
```

Out[19]: 'ometeam'

In [20]:
```
s1
```

Out[20]: 'welcometeam'

In [21]:
```
s1[12:]
```

Out[21]: ''

In [22]:
```
s1[13]
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[22], line 1
----> 1 s1[13]

IndexError: string index out of range
```

In [23]:
```
s1[:13]
```

Out[23]: 'welcometeam'

-s1[left:right] -s1[left:right:step]

In [24]:
```
s1
```

Out[24]: 'welcometeam'

In [25]:
```
s1[:15]
```

Out[25]: 'welcometeam'

In [26]:
```
s1
```

Out[26]: 'welcometeam'

In [27]:
```
s1[1:8:2]
```
Out[27]: 'ecmt'

In [28]:
```
s1
```
Out[28]: 'welcometeam'

In [30]:
```
s1[::3]
```
Out[30]: 'wcea'

In [29]:
```
s1
```
Out[29]: 'welcometeam'

In [34]:
```
s1[-9:-1:3]
```
Out[34]: 'lme'

In [30]:
```
s1
```
Out[30]: 'welcometeam'

In [37]:
```
s1[::-1]
```
Out[37]: 'maetemoclew'

In [31]:
```
s1
```
Out[31]: 'welcometeam'

In [39]:
```
s1[::-2]
```
Out[39]: 'meeolw'

string we are compeleted

25th NUMBER SYSTEM

In [40]:
```
25
```
Out[40]: 25

In [32]:
```
bin(25)  # binary base 2 = 1|0|0|1|1
```
Out[32]: '0b11001'

In [42]:
```
int(0b11001)  #    1.2^4+1.2^3+0+0+1.2^0 = 16+8+1 =25
```
Out[42]: 25

In [33]:
```
25
```
Out[33]: 25

In [44]:
```
oct(25)     # octal --- 3|1
```
Out[44]: '0o31'

In [45]:
```
int(0o31)   #  3.8^1+1.8^0 = 24+1 = 25
```
Out[45]: 25

In [46]:
```
hex(0)      #
```
Out[46]: '0x0'

In [48]:
```
hex(10)
```
Out[48]: '0xa'

In [49]:
```
hex(11)
```
Out[49]: '0xb'

In [50]:
```
hex(12)
```
Out[50]: '0xc'

In [51]:
```
hex(13)
```
Out[51]: '0xd'

In [52]:
```
hex(14)
```
Out[52]: '0xe'

In [53]:
```
hex(15)
```
Out[53]: '0xf'

In [54]:
```
hex(16)
```
Out[54]: '0x10'

In [56]:
```
int(0x10)  # 1.16^1+0 = 16
```
Out[56]: 16

BITWISE OPERATOR

In [57]:
```
# complement
~11
```

Out[57]: -12

In [58]:
```
~34
```
Out[58]: -35

In [59]:
```
~23
```
Out[59]: -24

and &

In [34]:
```
12 & 13          # 12 = 0b1100  13 = 0b1101   = 0b1100 =covert to number 12
```
Out[34]: 12

In [38]:
```
bin(12)
```
Out[38]: '0b1100'

In [62]:
```
bin(13)
```
Out[62]: '0b1101'

In [39]:
```
12 & 13 #bitwise and
```
Out[39]: 12

bitwise or

In [40]:
```
12 | 13 #bitwise or  12 = 0b1100  13 = 0b1101   = 0b1101 = convert into number 13
```
Out[40]: 13

xor ^

In [66]:
```
12 ^ 13   # 12 - 0b1100  ^ 13 - 0b1101 = 0b0001
```
Out[66]: 1

left shift <<

In [67]:
```
12 << 1     # gain a bit = 0b11000 = 16+8 = 24
```
Out[67]: 24

In [68]:
```
12 << 2   # gain a two bit = 0b110000 = 32+16 = 48
```

Out[68]: 48

right shift >>

In [1]:
```
12 >> 2   # loose a two bit = Ob11 = 2+1 = 3
```
Out[1]: 3

In [2]:
```
12 >> 3  # loose a three bit  = 0b1 = 1
```
Out[2]: 1

In [3]:
```
15 >> 2  # loose a two bit = 15 = 0b1111 = 0b11 = 2+1 = 3
```
Out[3]: 3

26th

In [4]:
```
x = sqrt(25)
```
```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[4], line 1
----> 1 x = sqrt(25)

NameError: name 'sqrt' is not defined
```

In [5]:
```
import math
```

In [6]:
```
math.sqrt(25)
```
Out[6]: 5.0

In [7]:
```
round(math.sqrt(25))
```
Out[7]: 5

In [8]:
```
pi = 3.14
pi
```
Out[8]: 3.14

In [7]:
```
pi = 4.17
pi
```
Out[7]: 4.17

In [9]:
```
math.floor(3.23)
```
Out[9]: 3

In [9]:
```
math.ceil(3.23)
```

Out[9]: 4

In [10]:
```
math.pow(3,2)
```

Out[10]: 9.0

In [11]:
```
m.pow(3,2)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[11], line 1
----> 1 m.pow(3,2)

NameError: name 'm' is not defined
```

In [12]:
```
import math as n
n.pow(3,2)
```

Out[12]: 9.0

In [13]:
```
import math as m

print(m.pow(3,2))
print(m.floor(3.2))
print(m.ceil(3.2))
```

```
9.0
3
4
```

In [18]:
```
from math import sqrt, pow, floor, ceil # math has many function if you want to

print(pow(2,3))
print(sqrt(10))
print(floor(2.3))
print(ceil(2.3))
```

```
8.0
3.1622776601683795
2
3
```

In [19]:
```
from math import * # math has many function if you want to import specific funct

print(pow(2,3))
print(sqrt(10))
print(floor(2.3))
print(ceil(2.3))
```

```
8.0
3.1622776601683795
2
3
```

input()

In [14]:
```
x = input()
x
```

Out[14]: '3'

In [15]:
```
x1 = input()
y1 = input()

z = x1 + y1
print(z)
```

54

In [16]:
```
type(y1)
```

Out[16]: str

In [26]:
```
x = int(input())
y = int(input())
z = x+y
print(z)
```

9

In [29]:
```
type(z)
```

Out[29]: int

In [17]:
```
x1 = int(input('Enter the 1st number')) #whenevery you works in input function i
y1 = int(input('Enter the 2nd number')) # it wont understand as arithmetic opera
z1 = x1 + y1
print(z1)
```

7

In [18]:
```
x1 = input('user_name') #whenevery you works in input function it always give yo
y1 = input('password') # it wont understand as arithmetic operator
z1 = x1 + y1
print(z1)
```

NARESHIT

In [19]:
```
st = input('enter a string')
print(st)
```

NARESH

In [20]:
```
print(st[2])
```

R

In [21]:
```
st = input('enter a string')[0]
print(st)
```

W

In [22]:
```
st = input('enter a string')[3:]
print(st)
```

SWORD

In [23]:
```
st = input('enter a string')
print(st)
```

10+20+5

In [ ]:
```python
st = int(input('enter a string'))
print(st)
```

In [43]:
```python
result = eval(input('enter an expr'))
print(result)
```

```
20
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [20]:
```python
#import keyword
#keyword.kwlist
```

In [ ]: